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Diffusion Models for Counterfactual Explanations: an Application to audio

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1 Introduction

Deep neural networks have achieved remarkable success across a wide variety of machine learning tasks, but their "black-box" nature remains a main problem for human trust in their outputs. As these models are increasingly deployed in real-world scenarios, the field of Explainable AI (XAI) has emerged to help humans understand the reasoning behind their decisions. Among the various XAI methods, counterfactual explanations are particularly intuitive. Rather than highlighting which features led to a specific decision, a counterfactual explanation answers the question: *"What minimal changes must be made to the input for the model to change its prediction?"*

Recently, the Diffusion Models for Counterfactual Explanations (DiME) framework [1] uses the state-of-the-art generative power of diffusion models to produce high-fidelity counterfactuals. By performing a partial classifier-guided diffusion, DiME successfully generates high-fidelity counterfactuals for facial attributes, ensuring a given image changes class while remaining perceptually close to the original input.

However, while XAI techniques have been widely studied in the field of computer vision (such as facial image processing), their application to the audio domain remains relatively unexplored. Audio signals have fundamentally different structural, temporal, and frequency dynamics compared to natural images, making the generation of coherent and realistic audio counterfactuals a unique and complex challenge. In this project, we aim to bridge this modality gap by adapting DiME to audio analysis, specifically focusing on multi-instrument recognition.

By converting 1D audio waveforms into 2D mel spectrograms, we maintain the core architecture of the DiME pipeline while addressing the unique specificities of the audio domain. This modality transition need significant changes, as the original dataset, the pre-trained diffusion model and the guidance classifier.

2 Diffusion Models for Counterfactual Explanations

Diffusion Models for Counterfactual Explanations (DiME) [1] takes advantage of the natural image generation power of diffusion models to produce counterfactual explanations for face images. Its objective is to invert a specific attribute in an initial image (for example changing a smiling face

to a non-smiling one) while preserving the original image's characteristics as much as possible. First, we will introduce relevant notions about diffusion models, before detailing the mechanism behind DiME.

2.1 Diffusion models in a nutshell

Diffusion models are a recent class of generative models that rely on two Markov chain sampling schemes that are inverses of one another. The basic idea is simple: by learning how images are progressively corrupted by noise in a forward pass, the model can reverse this process to generate new, natural images from pure noise.

The forward process The forward process starts from a clean natural image $x_0 \sim p(x)$ and iteratively samples $x_1, \ldots, x_T$ by progressively replacing the image with white Gaussian noise. At each timestep t, noise is added according to a given variance schedule β_t, following the transition distribution:

$$q(x_t|x_{t-1}) = \mathcal{N}(\sqrt{1-\beta_t}x_{t-1}, \beta_t I)$$

By defining $\alpha_t = 1 - \beta_t$ and their cumulative product $\bar{\alpha}_t = \prod_{k=1}^{t} \alpha_k$, this process can be simulated directly from the original clean image x_0 at any arbitrary timestep:

$$q(x_t|x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}x_0, (1-\bar{\alpha}_t)I)$$

As this process goes, the image becomes increasingly corrupted. In practice, the variances β_t are chosen such that after a large number of steps (often $T = 1000$), x_T is nearly indistinguishable from pure Gaussian noise, i.e. $x_T \sim \mathcal{N}(0, I)$. An example of this forward pass is illustrated in Figure 1 when read from right to left.

The reverse process In the reverse process, a neural network acts as a stochastic decoder, recurrently denoising x_T to recover the intermediate samples $x_{T-1}, \ldots, x_0$. The network takes the current timestep t and the noisy sample x_t as inputs to approximate the exact reverse transition, which is modeled as a Gaussian distribution:

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(\mu_\theta(x_t, t), \Sigma_\theta(x_t, t))$$

In practice, the network is trained to estimate the added noise $\epsilon_\theta(x_t, t)$, allowing it to compute the mean μ_θ. While the original Denoising Diffusion Probabilistic Models (DDPM) [2] formulation keeps the covariance Σ_θ fixed to a time-dependent constant, more recent methods allow the network to learn the variance as well, which is particularly useful for reducing sampling steps. By repeating this procedure, the model transforms random noise into a realistic clean image x_0 with zero variance (as seen in Figure 1 from left to right).

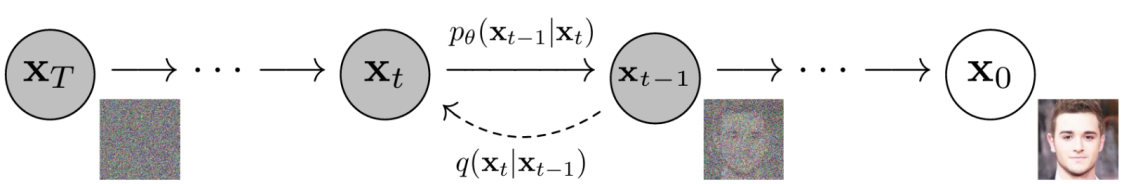

Fig. 1 Example of a forward pass (from right to left) and a backward pass (from left to right) in a diffusion model (from [2]).

Guided diffusion Furthermore, this sampling procedure can be adapted to optimize specific properties of the generated image using a scheme known as guided diffusion [3]. To condition the generation on external information, such as a given label y, the reverse step is shifted using the gradient of a loss function $L(x_t; y)$ that specifies the desired property:

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(\mu_\theta(x_t, t) - \Sigma_\theta(x_t, t)\nabla_{x_t} L(x_t; y), \Sigma_\theta(x_t, t))$$

This modification directly guides the denoising trajectory toward a specific attribute, making diffusion models highly effective for targeted image manipulations and counterfactual explanations.

2.2 How does DiME work?

First of all, note that DiME was introduced with different diffusion notations. The clean image is denoted x (instead of x_0), the noisy image at timestep t is denoted z_t (instead of x_t), and the cumulative product of the $1 - \beta_t$ is denoted α_t (instead of $\bar{\alpha}_t$). To align with the DiME framework formulation, we are going to use these notations while being careful when dealing with previous diffusion equations.

To generate a Counterfactual Explanation, the DiME framework first partially noises a clean image x to a timestep $\tau \in [\![1, T]\!]$ using the forward process, and then reverse the process using guided diffusion to move the image reconstruction towards a counterfactual, hence altered, version of the initial clean image. The loss used during the guided diffusion is composed of a classification term ($\mathcal{L}_{\text{class}}$) to guide the image reconstruction toward the target label, and a perceptual loss ($\mathcal{L}_{\text{perc}}$) to ensure proximity between the former image and its reconstruction. In the initial formulation of guided diffusion, the classifier was trained on noisy images and thus its gradient was informative along all the diffusion process. However, for lack of generality and because many classifiers are only trained over natural images, we could not extract information from the classification loss when dealing with noisy images z_t. That is why the idea of DiME is to fully reconstruct an intermediate clean image x_t (depending on z_t), that is used to compute clean gradients in order to guide the diffusion of z_t to z_{t-1}. This loop continues until having z_0, the counterfactual explanation of x without any remaining noise. The loss is thus expressed as:

$$\mathcal{L}(z_t; y, x) = \mathbb{E}\left[\lambda_c \mathcal{L}_{\text{class}}(C(y|x_t)) + \lambda_p \mathcal{L}_{\text{perc}}(x_t, x)\right] = \mathbb{E}\left[\tilde{\mathcal{L}}(x_t; y, x)\right]$$

where C is the considered classifier, y is the label we want to obtain, and thus $C(y|x_t)$ represents the posterior probability of the category y given x_t. Furthermore, λ_c and λ_p are constants that control the trade-off between classification guidance and perceptual proximity. It is important to note that an expectation is taken because x_t is stochastic with respect to z_t. In the original DiME article [1], the authors do not estimate the expectation as an empirical mean for computational reasons, but argue that a single realization per timestep t is sufficient as it will be averaged through the τ timesteps. Finally, they take advantage of this source of randomness to produce diverse counterfactual explanations from a single source image.

The x_t-based loss expression derives as follows:

$$\nabla_{z_t} \mathcal{L}(z_t; y, x) = \left(\frac{\partial x_t}{\partial z_t}\right)^T \nabla_{x_t} \tilde{\mathcal{L}}(x_t; y, x)$$

where the Jacobian $\frac{\partial x_t}{\partial z_t}$ would be computationally heavy when t is high. However, we can take advantage of the forward sampling process z_t from a clean image x_t, given by:

$$z_t = \sqrt{\alpha_t} x_t + \sqrt{1 - \alpha_t}\epsilon$$

to finally have a straightforward formulation of $\frac{\partial x_t}{\partial z_t}$ and thus derive:

$$\nabla_{z_t}\mathcal{L}(z_t; y, x) = \frac{1}{\sqrt{\alpha_t}}\nabla_{x_t}\tilde{\mathcal{L}}(x_t; y, x)$$

This whole pipeline is described by Figure 2.

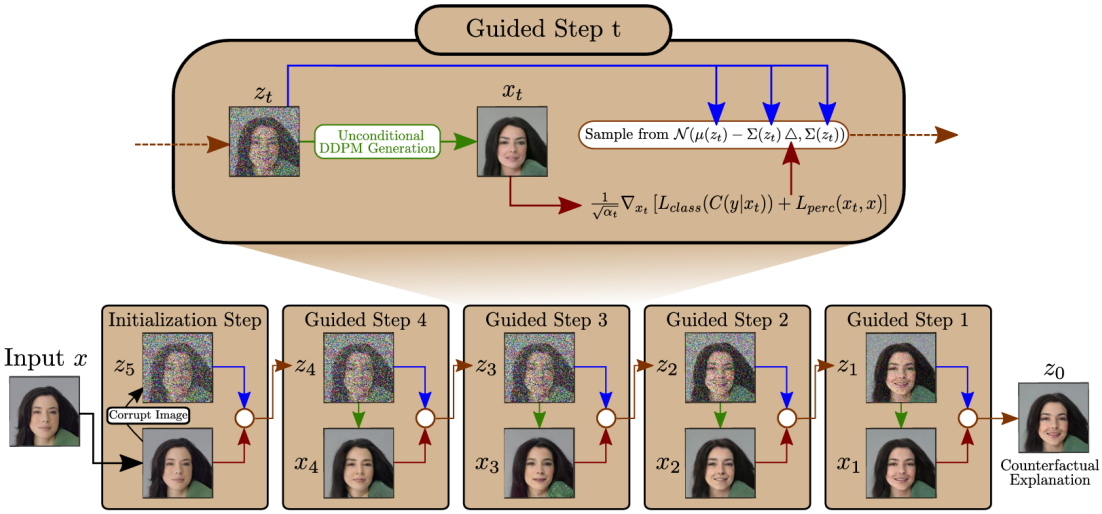

Fig. 2 DiME framework scheme with $\tau = 5$. At each diffusion step, we fully compute a clean image x_t from z_t using DDPM in order to have clean gradients to compute z_{t-1} using guided diffusion, to have an image with lesser noise than z_t and in the direction of being classified as y by the classifier.

A final important point is that if the counterfactual clean image z_0 is not classified as y by the classifier, the same pipeline is re-run with a higher λ_c constant, to give more importance to the class guidance in the loss expression. Finally, the method fails if it does not find any explanation after exhausting the provided values of λ_c.

3 Application of DiME to audio

This section aims to detail the implementation changes we made to adapt the initial DiME framework, applied on natural face images, to audio analysis, and more precisely classification from music files.

3.1 Dataset

The first and most important modification we propose is the data itself. While DiME initially uses the CelebA dataset [4], composed of human face images described by 40 binary attributes, our implementation is based on the OpenMIC-2018 dataset [5]. It contains 20000 examples of 10-seconds music, partially labeled for the presence or absence of 20 instrument classes by annotators on a crowd-sourcing platform.

3.2 Data transformation

Natural images are 2D spatial objects represented through 3 channels (Red, Green, Blue), while a raw audio is a 1D temporal signal. To bridge this dimensional gap, we can transform each

1D audio waveform into its mel spectrogram, a 2D time-frequency representation that maps the energy of the signal according to the mel scale, approximating human pitch perception. This means that, instead of using 3 channels images were each pixel represents a RGB color, we are going to use 1 channel images representing magnitude of the signal.

3.3 Pre-trained models

DDPM The original DiME implementation relies on a diffusion model trained to generate human faces, meaning its learned prior is incompatible with our audio data. Mel spectrograms are drawn from a completely different distribution and do not share the same features as natural images, thus we must adapt the backbone diffusion model. We opt for the DDPM published by Robert Dargavel Smith called "audio-diffusion-breaks-256"[1] and available on hugging face. It has been trained on 30000 samples from WhoSampled and YouTube, to generate mel spectrograms of size 256x256 corresponding to 5 seconds of audio.

Classifier Similarly, the guidance classifier $C(y|x_t)$ must be replaced to match the new domain. We keep the original DiME DenseNet121 architecture, but trained on 5 seconds spectrograms from OpenMIC-2018 instead of face images. This network provides the posterior probabilities for the target audio events, supplying the "clean gradients" necessary to guide the reverse diffusion process toward the desired audio counterfactual.

3.4 Perceptual loss

The perceptual loss ($\mathcal{L}_{\text{perc}}$) relies on a deep features extractor to ensure that the counterfactual image remains visually identifiable as the original input. In the DiME framework, the authors naturally used the standard VGG19 perceptual loss extracting visual features from the images. However, maintaining the visual similarity of spectrograms based on a natural image feature extractor does not strictly guarantee audio similarity. For this reason, we used the Pretrained Audio Neural Networks (PANNs) CNN-14, a feature extractor built specifically for handling spectrograms.

3.5 Metrics

Evaluating the realism and quality of the generated counterfactuals requires domain-appropriate metrics. While the original paper uses the Fréchet Inception Distance (FID) to measure image fidelity against the CelebA distribution, our audio implementation relies on the Fréchet Audio Distance (FAD). Instead of using visual embeddings, FAD utilizes audio embeddings (typically extracted via a pre-trained VGGish model) to compare the continuous distribution of the real audios from the dataset against our generated counterfactual spectrograms, providing a robust measure of acoustic plausibility. Finally, note that the counterfactual explanation of audio, and more precisely of music, has not been widely studied yet, so we cannot compare our results with other articles.

In addition to FAD, we report metrics that quantify (i) whether the desired classification change is achieved, and (ii) how strongly the counterfactual deviates from the original sample. Let $x^{(i)}$ be the original mel-spectrogram (treated as an image-like tensor), $\tilde{x}^{(i)}$ its counterfactual, $w^{(i)}$ and $\tilde{w}^{(i)}$ the corresponding time-domain waveforms, and $p_k^{(i)}$ (resp. $\tilde{p}_k^{(i)}$) the classifier posterior probability

[1] https://huggingface.co/teticio/audio-diffusion-breaks-256

for class k before (resp. after). We denote by y the target class and by τ the decision threshold (we use $\tau = 0.5$). All reported values are averaged over N samples.

Flip ratio (%). This measures the fraction of samples for which the classifier decision for the *target* class changes between the original and the counterfactual: $\text{Flip}(\%) = 100 \cdot \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\left[\mathbf{1}\left(p_y^{(i)} > \tau\right) \neq \mathbf{1}\left(\tilde{p}_y^{(i)} > \tau\right)\right]$. A high flip ratio indicates that counterfactuals achieve the intended class change.

ℓ_1 **proximity (L1 Prox).** We quantify how close the counterfactual spectrogram remains to the original in pixel space: $\text{L1 Prox} = \frac{1}{N}\sum_{i=1}^{N} \left\|\tilde{x}^{(i)} - x^{(i)}\right\|_1$, (optionally normalized by the number of pixels HW if needed).

MNAC (Mean Number of Attributes Changed). Because our classifier is multi-label, it is important to check whether non-target instruments are unintentionally modified. MNAC counts how many *non-target* labels change their binary decision after thresholding: $\text{MNAC} = \frac{1}{N}\sum_{i=1}^{N} \sum_{k \neq y} \mathbf{1}\left[\mathbf{1}\left(p_k^{(i)} > \tau\right) \neq \mathbf{1}\left(\tilde{p}_k^{(i)} > \tau\right)\right]$. Lower MNAC means more class-specific counterfactual changes.

SNR (dB). In the time domain, we measure a signal-to-noise ratio between the original waveform and the counterfactual: $\text{SNR(dB)} = 10\log_{10}\left(\frac{\frac{1}{T}\sum_{t=1}^{T}\left(w_t^{(i)}\right)^2 + \varepsilon}{\frac{1}{T}\sum_{t=1}^{T}\left(\tilde{w}_t^{(i)} - w_t^{(i)}\right)^2 + \varepsilon}\right)$, with a small constant $\varepsilon > 0$ for numerical stability. Higher SNR indicates stronger waveform preservation.

LSD (Log-Spectral Distance). We also report a spectral distortion measure based on STFT magnitudes. Let $S_w(f,m)$ denote the STFT magnitude at frequency bin f and time frame m. Then: $\text{LSD} = \frac{1}{M}\sum_{m=1}^{M} \sqrt{\frac{1}{F}\sum_{f=1}^{F}\left(20\log_{10}(|S_w(f,m)| + \varepsilon) - 20\log_{10}(|S_{\tilde{w}}(f,m)| + \varepsilon)\right)^2}$. Lower LSD indicates closer spectra.

Mean cosine similarity (embedding-based source preservation). To better reflect perceptual similarity beyond sample-wise distances, we compute the cosine similarity between embeddings extracted from a pre-trained audio network $f(\cdot)$: $\text{CosineSim} = \frac{1}{N}\sum_{i=1}^{N} \frac{\left\langle f(w^{(i)}), f(\tilde{w}^{(i)})\right\rangle}{\left\|f(w^{(i)})\right\|_2 \left\|f(\tilde{w}^{(i)})\right\|_2}$. Higher values indicate better preservation of the original content in the embedding space.

FAD (Fréchet Audio Distance). Finally, we compute FAD by comparing the distribution of embeddings of reference audios and counterfactual audios. If (μ, Σ) and $(\tilde{\mu}, \tilde{\Sigma})$ are the empirical mean and covariance of the reference and counterfactual embedding sets, respectively, then: $\text{FAD} = \|\mu - \tilde{\mu}\|_2^2 + \text{Tr}\left(\Sigma + \tilde{\Sigma} - 2\left(\Sigma\tilde{\Sigma}\right)^{1/2}\right)$. Lower FAD indicates that counterfactuals are closer to the reference audio distribution in the chosen embedding space (e.g. VGGish).

3.6 Overview of our experiment

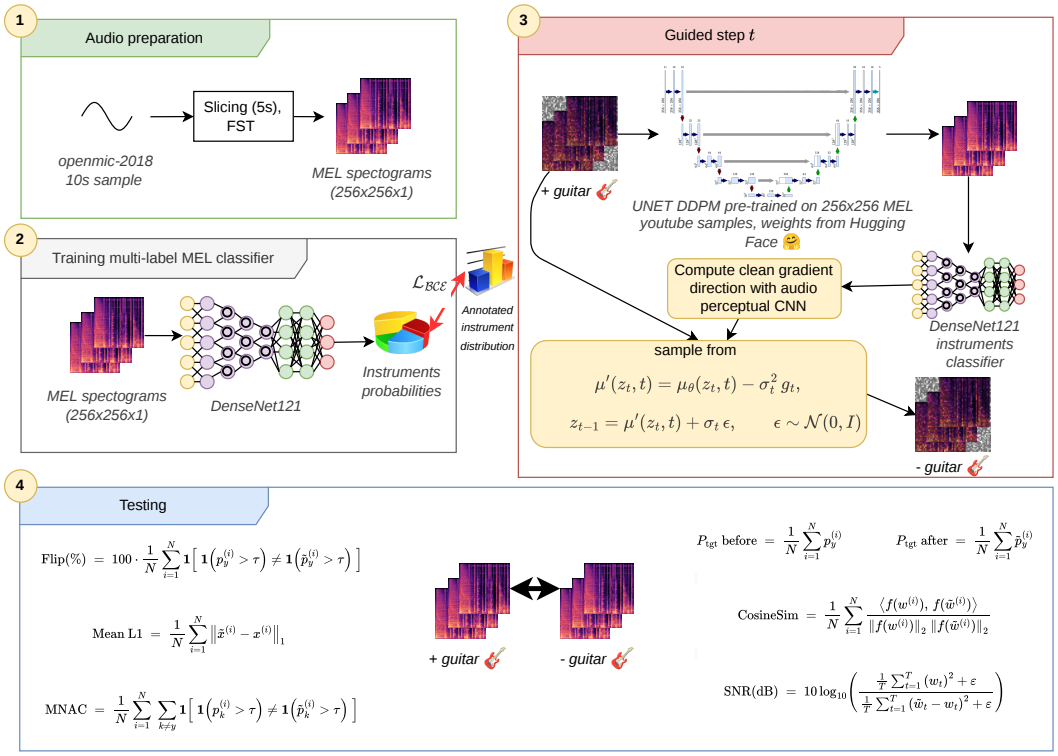

Fig. 3 Scheme of our audio adaptation of the DiME framework. The main differences with the original DiME are the data (from face images to music spectrograms), the pre-trained models (from a DDPM trained on faces and a classifier trained on CelebA to a DDPM trained on spectrograms and a classifier trained on OpenMIC), and the perceptual loss (from VGG19-based to CNN14-based). First, we transform the audio samples to MEL spectrograms of 5seconds (openmic-2018 samples), then we train a classifier on these spectograms to predict the presence of the 20 instruments. Finally, we use the pre-trained DDPM to generate counterfactual spectrograms with the DiME pipeline, and we evaluate them using the metrics described in subsection 3.5

4 Results

Given an input spectrogram x, we sample a partially noised version $z_\tau \sim q(z_\tau \mid x)$ at timestep $\tau = 160$ and run the reverse diffusion process to obtain a counterfactual $\tilde{x}$. Guidance is computed from a multi-label DenseNet121 classifier (trained on OpenMIC) together with proximity regularization, combining an ℓ_1 term (weight $\lambda_{\ell_1} = 0.02$) and a CNN14-based perceptual term (weight $\lambda_p = 20$). For classifier guidance, we use a scale $\lambda_c = 100$.

In the `guitar_removal_hq` run, targets are specified explicitly rather than sampled from pseudo-labels: we provide a sequence of target class indices with `-target_labels` and set the desired direction to removal ($y_{\text{val}} = 0$), producing sequential counterfactual steps saved as `step_0`, `step_1`. Moreover, we apply guidance only during the first 20 reverse steps (parameter `guided_iterations=20`); the remaining denoising steps follow the unguided reverse diffusion dynamics.

4.1 Qualitative results

We first provide qualitative evidence that the proposed guided diffusion procedure can produce counterfactual mel-spectrograms that remain close to the input while introducing structured, localized changes. Figure 4 shows (i) the original spectrogram, (ii) the generated counterfactual at Step 0, and (iii) an absolute difference map. The difference map suggests that the edit is not a global corruption of the spectrogram, but rather a sparse modification pattern, which is consistent with the relatively small ℓ_1 proximity values reported in Table 1.

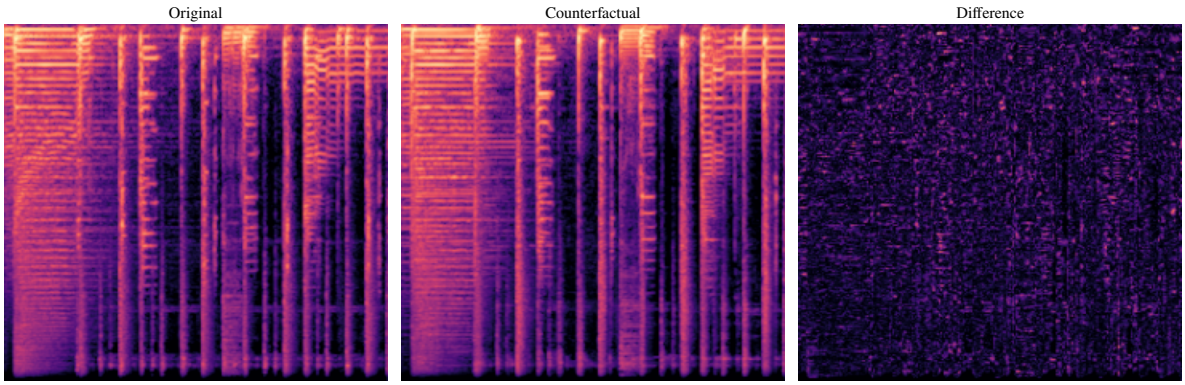

Fig. 4 Qualitative comparison (`guitar_removal_hq`, Step 0, sample 0): original mel-spectrogram (left), counterfactual (middle), and absolute difference map (right).

We also listen to the reconstructed waveforms corresponding to the original and counterfactual spectrograms. The original sample contains a guitar, while the counterfactual is intended to remove it. Subjectively, we can confirm that the guitar is less audible in the counterfactual, while other instruments are still present. The samples are available here: original and counterfactual.

4.2 Quantitative results

We report quantitative results for the sequential counterfactual experiment `guitar_removal_hq`. Table 1 summarizes the main metrics per step (flip ratio, proximity, and waveform-based similarity). In addition, Table 2 reports target confidence statistics before and after the edit.

Table 1 Main quantitative results for `guitar_removal_hq`. Flip Ratio measures success for the target edit; L1 Prox measures spectrogram proximity; MNAC counts non-target label changes. SNR and LSD are computed on reconstructed waveforms.

experiment	step	Flip Ratio (%)	L1 Prox	MNAC	SNR (dB)	LSD
guitar_removal_hq	0	100.000000	0.110702	1.800000	-2.950137	9.364269
	1	62.500000	0.106687	3.000000	-3.073757	9.210129
	2	100.000000	0.107320	2.375000	-2.981012	9.261023
	3	100.000000	0.106692	2.125000	-2.908162	9.258429
	4	100.000000	0.105072	1.875000	-3.239836	9.221994
	5	100.000000	0.104598	1.375000	-3.153058	9.172865

Table 2 Per-step target-class evaluation for `guitar_removal_hq`. P_{tgt} denotes the classifier posterior probability for the target class before and after counterfactual generation.

Step	Samples	Flip (%)	Mean L1	MNAC	P_{tgt} before	P_{tgt} after
0	10	100.00	0.110702	1.8000	0.1670	0.0996
1	8	62.50	0.106687	3.0000	0.8218	0.3656
2	8	100.00	0.107320	2.3750	0.6552	0.0963
3	8	100.00	0.106692	2.1250	0.6480	0.1863
4	8	100.00	0.105072	1.8750	0.3103	0.0619
5	8	100.00	0.104598	1.3750	0.5214	0.1049

Overall, flip ratios are high across steps (often reaching 100%), showing that classifier guidance effectively enforces the requested edit. Step 1 is more challenging (flip ratio 62.5%) and also exhibits the largest MNAC, suggesting that achieving the target flip at that stage may induce more side effects on other predicted instruments. Across steps, the mean ℓ_1 proximity remains relatively stable (around 0.10–0.11), indicating that counterfactuals remain close in spectrogram space.

Waveform-based distortion metrics (SNR, LSD) are substantially degraded. In our setting, these metrics depend on the spectrogram-to-audio inversion procedure (used to reconstruct waveforms), which can introduce artifacts. We therefore also report embedding-based similarity and distributional metrics computed directly on available waveform pairs.

Table 3 Embedding-based similarity and realism metrics computed on waveform pairs available for evaluation. Cosine similarity is computed between embeddings of original and counterfactual waveforms; FAD compares the embedding distributions of original vs. counterfactual audio.

Pairs	Mean cosine similarity	Embedding model
8	0.9772	microsoft/unispeech-sat-base-plus-sv

Embedding model	FAD
vggish	2.7064

4.3 Discussion

The quantitative results confirm the intended behavior of our guided diffusion pipeline: the target confidence P_{tgt} consistently decreases after counterfactual generation when performing a removal edit, and the flip ratio is high for most steps (Table 2). At the same time, MNAC shows that non-target predictions can change as a side effect, especially in the most difficult steps (e.g., Step 1). This reflects the inherent entanglement of instrument cues in music spectrograms and the fact that classifier gradients are not guaranteed to be class-disentangled.

From a fidelity perspective, the relatively small L1 proximity values suggest that edits remain limited in spectrogram space (Table 1), and the embedding cosine similarity remains high (Table 3), indicating that much of the global audio content is preserved in a learned audio representation. However, the non-zero FAD score highlights that counterfactuals are still detectably different from the reference audio distribution.

Finally, the discrepancy between spectrogram-space proximity and waveform-based distortion metrics suggests that waveform reconstruction can be a bottleneck for audio evaluation in our setting. This motivates the use of embedding-based metrics (similarity, FAD) and points toward

future improvements such as better inversion/vocoder components or end-to-end evaluation directly in the spectrogram domain.

5 Limitations and difficulties

Initially, the goal of this project was to reimplement a portion of DiME, as its code is not explicitly linked in the paper and the instruction given in the list of papers was not about changing the modality. However, after further research, we found the paper's official GitHub repository[2] containing the complete implementation. Thus, we decided to change our approach and apply the framework to a completely new modality that is the audio.

The primary challenge we encountered was not the structural shift to 2D audio representations, but rather the lack of available pre-trained models for our target domain. Initially, we wanted to adapt DiME to the FSD50K dataset [6], composed of diverse environmental sound events annotated with 200 binary attributes such as "Male speech" or "Wind". However, we were constrained by the absence of a robust and open-source DDPM trained on general background noise, and training our own was out of the scope of this project. Because the only suitable pre-trained diffusion model we found was trained specifically on music spectrograms, we finally adapted our pipeline from general sound classification to music-specific tasks. Furthermore, we could not find a suitable pre-trained classifier on this new domain, so we ended-up by training our own from scratch. As expected, combining a niche DDPM with a custom classifier led to poorer results than the original DiME framework, based on state-of-the-art models in the heavily studied area that is the face analysis.

[2] https://github.com/guillaumejs2403/DiME

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